# Causal Forecast-Conditioned BS-RoFormer for Ocean Sound Separation

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***Abstract*—We ask whether a strictly causal estimate of the expected ocean background can help separate an unexpected acoustic event. FC-BSR forecasts the current one-third-octave spectrum from preceding soundscape history and injects it into BS-RoFormer through concatenation, feature-wise linear modulation (FiLM), or cross-attention. A three-tier evaluation separates mechanism selection from field realism. With exact synthetic stems, FiLM improves SI-SDR improvement (SI-SDRi) by 0.34 dB over an unconditioned separator. With 42 h of continuous hydrophone background and injected targets, causal priors improve all 5-s to 10-min resolutions; the 5-s setting gains 1.226 dB, although persistence is the best forecaster at three resolutions. On 1,225 annotated real whale calls without isolated stems, FC-BSR reduces residual event probability from .523 to .507 relative to matched BS-RoFormer, while slightly increasing off-event change. The results support forecast conditioning, but not an unconditional fidelity or deployment claim.**

***Index Terms*—ocean acoustics, source separation, soundscape forecasting, BS-RoFormer, weak-reference evaluation**

## I. Introduction

Ocean backgrounds vary with shipping, weather, biology, and season [1]. Conventional separators infer the target and this changing background from the same mixture. We instead provide a causal prior: the spectrum expected from observations ending before the mixture arrives. The scientific question is

$$S_{t-H:t-1} \xrightarrow{P_\psi} \widehat{S}_t, \qquad x_t = N_t + \alpha s_t \xrightarrow{F_\phi(\cdot|\widehat{S}_t)} (\hat{s}_t, \hat{N}_t). \quad (1)$$

The model never observes the current clean background $N_t$.

Time-domain separators such as Conv-TasNet and DPRNN [2], [3], attention models such as SepFormer and TF-GridNet [4], [5], and band-split models [6], [7] provide strong mixture-only backbones. Hybrid Transformers offer another competitive design [8], while MixIT reduces dependence on isolated sources [9]. None tests a forecast of the upcoming acoustic background as side information. Conditional modulation [10], cross-attention [11], and cross-modal separation [12] suggest three plausible interfaces. Meanwhile, DeepAR, N-BEATS, DLinear, and PatchTST span recurrent, basis, linear, and patch-based forecasting [13]–[16].

Our contributions are: (1) a causal forecast-conditioned BS-RoFormer with a matched three-way conditioning ablation; (2) a leakage-audited evaluation that progresses from complete-reference surrogates to real background and finally real events; (3) tests across signal strength, source diversity, and 5-s, 30-s, 1-min, and 10-min resolutions; and (4) a reference-free real-event test that directly compares FC-BSR with unconditioned BS-RoFormer.

## II. Method

### A. Forecast and separator

Historical one-third-octave levels (TOL) $S_{t-H:t-1} \in \mathbb{R}^{H\times F}$ are normalized using training data only. A residual GRU predicts

$$z_t = \mathrm{LN}(\mathrm{GRU}(S_{t-H:t-1})), \quad \widehat{S}_t = S_{t-1} + g_\psi(z_t). \quad (2)$$

Residual initialization exactly represents persistence. Learned forecasting is retained only when validation MAE improves on that baseline, consistent with proper point-forecast comparison [17].

BS-RoFormer maps the mixture STFT to band tokens $X_k$. At the same pre-axial- Transformer location, we compare

$$\begin{aligned} \text{Concat: } \widetilde{X}_k &= X_k + f([X_k; P_k(\widehat{S}_t)]), \\ \text{FiLM: } \widetilde{X}_k &= (1+\tanh\gamma_k)\odot X_k + \beta_k, \\ \text{X-attn: } \widetilde{X} &= X + \mathrm{Attn}(X, E(\widehat{S}_t), E(\widehat{S}_t)). \end{aligned} \quad (3)$$

Residual branches start at zero so all models share the same initial audio mapping. The loss combines waveform $L_1$, multi-resolution complex-STFT loss, and, in the surrogate study only, $0.2\|\widehat{S}_t - S_t\|_1$. In real-waveform studies the validation-selected predictor is frozen before separator training.

### B. Data and evidence ladder

Tier 1 uses NOAA–Navy SanctSound hourly TOL products [18]. MB03 deployments 01–02/03/04 are train/validation/test. Random-phase inversion yields only a controlled spectral surrogate, not recovered audio. Targets include harmonics, chirps, inharmonic and FM tones, impulses, and band noise. Spectral shift uses Jensen–Shannon (JS) divergence [19]; entropy separately measures diffuseness.

Tier 2 uses 42 h of continuous 8-kHz hydrophone WAV: 12/6/24 h for train/ validation/test. One-second periodograms are aggregated into 23 TOL bands and 5-s, 30-s, 1-min, or 10-min windows. The next real waveform $N_t$ receives a synthetic target, preserving exact references. Each resolution covers one hour of causal history.

Tier 3 uses day-disjoint, 2-kHz DCLDE 2013 right-whale recordings [20]. Strong event annotation supports detection [21], but no isolated event/background stems exist. Clean training windows receive synthetic upcalls; 1,225 untouched real positive test clips evaluate transfer. Both arms share samples, seeds, initialization, 2,500-update budget, and validation-only checkpoint selection.

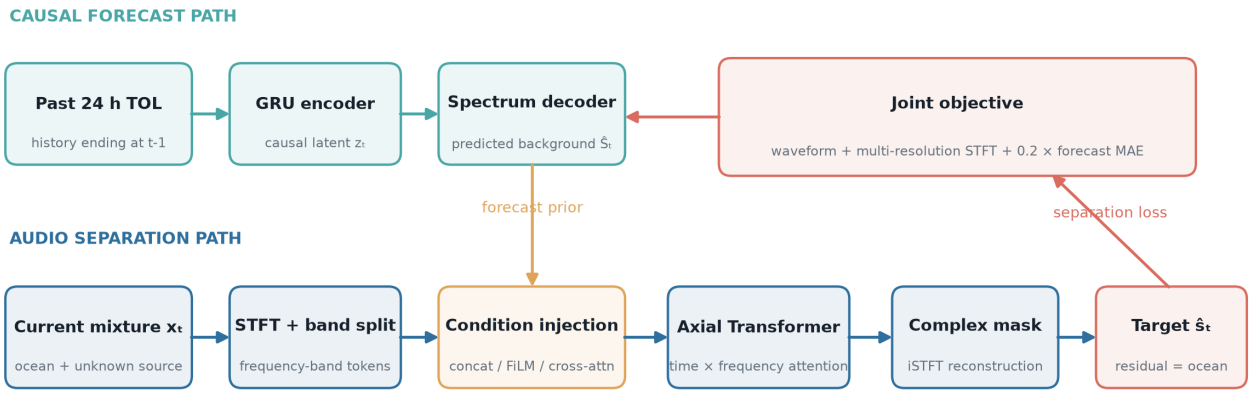



Fig. 1. FC-BSR. Causal history forecasts the current background spectrum; the prior enters the same BS-RoFormer backbone by concatenation, FiLM, or cross-attention. The baseline removes the blue forecast path.

### C. Protocol and metrics

Every comparison is paired at the waveform level. Mixture time, target draw, SNR, random seed, separator-core initialization, optimizer schedule, and checkpoint rule are identical across conditioned and unconditioned arms. Deployment 03 selects all checkpoints; deployment 04 is evaluated once. The continuous-audio split is chronological and DCLDE is split by recording day, so overlapping windows cannot cross partitions. Feature normalization is fit only on training recordings, and the historical encoder ends at $t-1$.

For complete-reference tiers we report SI-SDR and improvement over the input mixture, residual-background SI-SDR, forecast MAE in dB, parameter count, and real-time factor (RTF). SI-SDR follows scale-invariant projection rather than the original BSS Eval decomposition [22], [23]. Uncertainty is a seeded paired bootstrap over test clips. Diversity tests fix input SNR at 0 dB and pair 128 examples from each of six source families, preventing strength from confounding source shape. We report both directional $D_{\mathrm{KL}}(p\|q)$ and finite symmetric JS divergence, but interpret them as spectral displacement rather than semantic diversity.

## III. Experiments and Results

### A. Controlled selection and diversity

All Tier-1 variants use 4,000 examples per training deployment, 2,000 updates, and SNR sampled from $[-12, 12]$ dB. Table I selects FiLM on validation; its paired 0-dB gain is 0.34 dB (95% bootstrap CI $[0.21, 0.48]$). Band-wise scale and bias efficiently express "expected energy here," whereas concatenation must compress doubled features and cross-attention must infer alignment from a coarse spectrum.

At SNR $-12, -6, 0, +6, +12$ dB, FiLM obtains SI-SDRi $7.30, 6.95, 5.81, 3.51, -0.10$ dB; absolute SI-SDR at $+12$ dB remains 11.94 dB. On 768 paired, fixed-0-dB examples, FiLM improves 5.36 to 5.70 dB. Gain has weak correlation with JS shift ($-0.085$) but positive correlation with spectral entropy ($0.145$); the highest-entropy quintile gains 0.80 dB. Thus KL/JS distance broadens coverage, but does not itself predict when the prior helps. Fig. 2 visualizes the causal history, current spectrum, mixture, estimates, and spectra for a diffuse source.

TABLE I
Matched conditioning ablation (dB).

| Injection | Val. SI-SDRi | Test SI-SDRi | Paired $\Delta_0$ |
|---|---|---|---|
| None | 4.96 | 5.00 | 0 |
| Concatenation | 5.17 | 5.30 | +0.15 |
| FiLM | **5.35** | **5.39** | **+0.34** |
| Cross-attention | 4.99 | 5.11 | +0.01 |

The family breakdown supports that interpretation: band-limited noise gains 0.54 dB ($[0.19, 0.92]$), impulse trains 0.47 dB ($[0.16, 0.79]$), and FM targets 0.43 dB ($[0.08, 0.82]$). By contrast, gain falls from 0.54 dB in the lowest JS quintile to 0.00 dB in the highest. A larger divergence therefore expands the tested support but does not guarantee benefit; overlap with predictable ocean bands and within-source diffuseness are more plausible explanatory variables.

### B. Continuous real background

Predictors train for at most 2,500 updates; separators train for 3,000–5,000. Table II reports 1,024 paired held-out clips per resolution. Every confidence interval excludes zero, but overlaps the other resolutions. Only 5 s selects a learned GRU (1.4% lower MAE than persistence); 30 s, 1 min, and 10 min select persistence. Temporal context is useful, but current data do not show that complex dynamics are generally necessary.

Validation MAE is 1.366, 0.782, 0.687, and 1.371 dB from 5 s to 10 min. The 10-min training set contains only 60 distinct next-window targets, explaining its strong forecast overfitting. All four separators pass the five-SNR engineering gate and run at RTF $\approx .0024$. Thus the main positive result is the value of time-correlated conditioning, not a statistically established

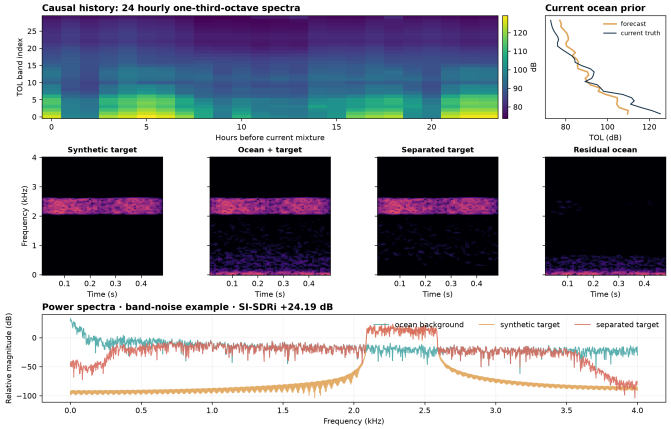



Fig. 2. Representative high-entropy target: historical/predicted spectrum, mixture and separated spectrograms, and spectral overlays.

TABLE II
REAL-BACKGROUND FULL-REFERENCE RESULTS.

| Window | Forecast | Cond. | Base | Gain [95% CI] |
|---|---|---|---|---|
| 5 s | GRU | 29.753 | 28.527 | 1.226 [1.082,1.381] |
| 30 s | Persist. | 29.630 | 28.534 | 1.095 [0.935,1.265] |
| 1 min | Persist. | 29.661 | 28.533 | 1.128 [0.971,1.298] |
| 10 min | Persist. | 29.711 | 28.535 | 1.176 [1.036,1.326] |

Cond./Base/Gain are SI-SDRi in dB.

ranking of temporal resolution or universal superiority of the GRU.

### C. Real events without separated ground truth

Without isolated stems, full-reference metrics are unavailable. Following learned referenceless metrics [24], [25] and semantic audio embeddings [26], [27], we train, calibrate, and freeze an event classifier (test AUROC .9704). Mixture consistency enforces $\hat{N} = x - \hat{s}$. We then restore $\hat{s}$ at gains $-60$ to 0 dB and require event probability to recover monotonically. Identity and all-to-event controls reject trivial solutions; off-event relative change audits preservation.

The balanced classifier manifest contains 5,292/2,696/2,676 train/validation/ test clips; certain upcalls define positives, while negatives exclude certain and possible calls with a 2-s margin. Test average precision is .9738 and F1 is .9196. Each eligible target receives eight preceding 8-s, 30-band spectra. The residual GRU lowers validation forecast MAE from 2.081 to 1.921 dB, then is frozen. Controlled validation SI-SDRi is 13.871 dB for BS-RoFormer and 13.925 dB for FC-BSR before either model sees the real-event test set.

FC-BSR lowers residual event probability by .0160; a paired bootstrap over three recording-day means gives $[-.0210, -.0132]$. Event-stem probability and monotonicity are unchanged within uncertainty, but off-event change increases by .00662 ($[.00331, .00781]$). Hence forecast conditioning improves semantic removal beyond matched BS-RoFormer, with a small preservation cost.

TABLE III
MATCHED COMPARISON ON 1,225 REAL CALLS.

| Method | $p(\hat{N})\downarrow$ | $p(\hat{s})\uparrow$ | Mono.↑ | Off-event↓ |
|---|---|---|---|---|
| BS-RoFormer | .523 | .901 | .986 | **.0956** |
| FC-BSR | **.507** | .901 | .985 | .1022 |

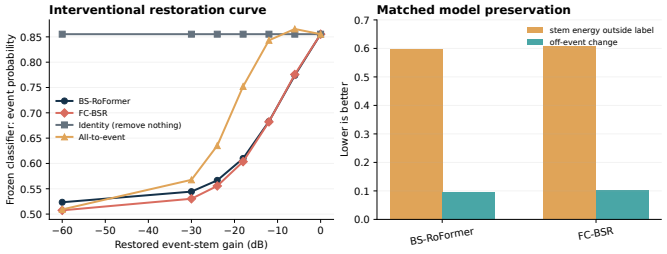



Fig. 3. Reference-free restoration curves (left) and preservation audit (right). FC-BSR suppresses events more strongly but changes more off-event audio.

The restoration gains are $\{-60, -30, -24, -18, -12, -6, 0\}$ dB. Identity ($\hat{s} = 0$) retains the original event probability and all-to-event ($\hat{s} = x$) destroys the background, showing why no single classifier score is sufficient. Event-stem probability changes by only $-.00035$ ($[-.00283, .00552]$) and monotonicity by $-.00109$ ($[-.00342, .00265]$), whereas energy outside annotated intervals increases by .0101 ($[.00310, .01574]$). These partial-ground-truth checks rule out obvious semantic gaming but cannot identify phase or waveform distortion inside an event interval.

## IV. DISCUSSION AND CONCLUSION

The evidence supports nested claims. Controlled stems select FiLM; real background confirms the causal prior across four sub-hour resolutions; real events show a semantic advantage over unconditioned BS-RoFormer. It does not show universal benefit from learned forecasting: persistence wins three of four resolutions. Nor can classifier scores recover unavailable waveform fidelity. Multi-site, multi-season recordings, more 10-min targets, multiple seeds, and isolated or stronger partial references remain necessary.

This ordering also limits interpretation. Tier 1 identifies the conditioning mechanism but uses phase-synthesized backgrounds. Tier 2 removes that assumption while retaining synthetic targets and exact stems. Tier 3 removes the synthetic-test-target assumption but necessarily loses full-reference fidelity. Agreement across tiers is stronger than any one result, yet it does not make their metrics interchangeable. In particular, classifier-based event removal is evidence of semantic transfer, not a replacement for SI-SDR.

Practical gates are met: 5 dB overall SI-SDRi, 3 dB improvement for SNR $\leq 0$ dB, 8 dB absolute SI-SDR for stronger inputs, and RTF below one. A local frontend visualizes the forecast and plays/downloads both estimates; its 10-s chunked RTF is .043 on an RTX 3060. This supports inspection, not field deployment.

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